

flea

NEWS62

June 2008



<http://www.entsoc.org/networks/Interest/fleanews.htm>

FLEA NEWS is a biannual newsletter about fleas (Siphonaptera). It is mainly bibliographic in nature, but recipients are urged to check any citations given here before including them in publications. Many of our sources are abstracting journals and current literature sources such as Current Contents(R), National Agricultural Library (NAL) Agricola, and National Library of Medicine (NLM) Medline, and citations have not necessarily been checked for accuracy or consistent formatting.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed in Agricola or Medline databases, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book reviews, Biography, Hypotheses and Literature Reviews, and Anecdotes. Send to:

R. L. Bossard, Ph.D.
 Editor, Flea News Biology Dept.
 Westminster College
 1840 South 1300 East
 Salt Lake City, UT 84105 USA
rbossard@westminstercollege.edu

Organizers of the Flea News Network are Drs. R. L. Bossard and N. C. Hinkle.

N. C. Hinkle, Ph.D.
 Dept. of Entomology
 Univ. of Georgia
 Athens GA 30602-2603 USA
NHinkle@uga.edu
 (706) 583-8043

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Announcements

Flea News is archived permanently on the internet, thanks to the Entomological Society of America (ESA). The ESA sponsors Networks related to entomology, but you do not need to be a member to access Flea News. The Flea News archive is currently at:

<http://www.entsoc.org/networks/Interest/fleanews.htm>

The Handbook of British Fleas, first published by Frans G.A.M. Smit in 1957, has now been updated. The reference is:

Whitaker, A.P., 2007. Fleas (Siphonaptera). Handbooks for the Identification of British Insects. Vol. I Part 16 (2nd Ed.) 178pp. (ISBN 978 0 90154 685 2)

In addition, distribution data contained within it has also been put on the web, as a searchable database:

Whitaker, A.P., 2007. The Distribution of British Fleas Database
URL: <http://www.nhm.ac.uk/research-curation/projects/british-flea-distribution/index.html>

R.S. George. 2008. Atlas of the Fleas (Siphonaptera) of Britain and Ireland.
ISBN 978-0-95576-723-4, 98 pp, £8.00. Ordering Information: CEH Publication Sales, Monks Wood, Abbots Ripton, Huntingdon, Cambs, PE28 2LS.

The World Health Organization has released a book entitled "Public Health Significance of Urban Pests". A pdf of the entire book is available on-line at

<http://www.euro.who.int/document/e91435.pdf> , or you can order a hard copy for \$120 plus \$12 shipping (http://www.euro.who.int/InformationSources/Publications/Catalogue/20080617_9).

Topics covered include allergens, cockroaches, house dust mites, bedbugs, fleas, pharaoh and fire ants, flies (by Jerry Hogsette), birds, human body lice, ticks, mosquitoes, commensal rodents and other vertebrate pests. One useful aspect is that the authors attempt to provide estimates of economic value of the various pests and their suppression (at least for Europe and North America).

New and updated:

Physician's guide to arthropods of medical importance, 5th edition by J. Goddard (Mississippi State Dept. of Health), CRC Press. ISBN: 978 0 8493 9539 1. \$159.95 / £88.00.

Robert E. Lewis was outstanding as editor of Flea News, but he retired in July 2000. Robert Bossard and Nancy Hinkle are now organizing the Flea News Network. They hope Flea News redux will be a useful publication to Siphonapterists. We thank all who have sent items for Flea News.

Meetings

The XXIII International Congress of Entomology, Breaking the Barriers, is slated for July 6-12, 2008, in Durban, South Africa. This event is sponsored by the Entomological Society of South Africa. URL: <http://www.ice2008.org.za>

The 41st Annual Meeting of the Society for Invertebrate Pathology and 9th International Conference on *Bacillus thuringiensis* will be held August 3-7, 2008 at the University of Warwick, UK. URL: <http://www.ent.iastate.edu/sip/2008>.

The 93rd Annual Meeting of the Ecological Society of America is slated for August 3-8, 2008, in Milwaukee, Wisconsin. URL: <http://www.esa.org>

The 2008 Annual International Conference on Diseases in Nature Communicable to Man will be held in Regina, Saskatchewan from August 10-12, 2008. <http://incdncm.googlepages.com/PestSummit2008>, at Sofitel for August 13-15, 2008. Centara Grand, Bangkok, Thailand. Contact Nattaphong, nattaphong.n@issworld.com

The CONFERENCE of the Society of Vector Ecology (SOVE) will be in Fort Collins, Colorado September 28th - October 2nd, 2008. <http://www.sove.org/Home.html>

Second International Conference on Dengue and Dengue Haemorrhagic Fever, will be held October 15-17, 2008 at Hilton Phuket Arcadia Resort and Spa, Phuket Island, Thailand, www.dengue2008.phuket.com

The 2008 Entomological Society of America Annual Meeting will be held on November 16-19, 2008, at the Reno-Sparks Convention Center in Reno, Nevada. Contact: ESA Annual Meeting, 10001 Derekwood Lane, Suite 100, Lanham, MD 20706, phone 301-731-4535, fax 301-731-4538

Directory of Siphonapterists

Dr. J.C. Beaucournu
Parasitologie médicale
Faculté de Médecine de Rennes
2, avenue du Professeur Léon Bernard
F 35043 Rennes cedex
France
jeanclaude.beaucournu@gmail.com

Dr. Marian Blaski
Silesian University, Department of Zoology
ul. Bankowa 9
40-007 Katowice
Poland
marian.blaski@us.edu.pl

Dr. R.L. Bossard
Biology Dept.
Westminster Coll.
Salt Lake City, UT 84105
rbossard@westminstercollege.edu
ecology of host-parasite relationships.

Dr. Michael Dryden
E.J. Frick Professor of Veterinary Medicine
Department of Diagnostic Medicine/Pathobiology
Coles Hall
Manhattan, KS 66506
Dryden@vet.ksu.edu
veterinary parasitology.

Dr. Terry Galloway
Professor of Entomology and Associate Curator
J.B. Wallis Museum of Department of Entomology
Winnipeg, Manitoba
Canada R3T 2N2
Terry_Galloway@umanitoba.ca
ecology and taxonomy, especially of the larvae.

Dra. M^a Soledad Gómez
 Laboratori de Parasitologia
 Facultat de Farmacia, Universitat de Barcelona
 Avda. Joan XXIII s/n, 08028
 Barcelona
 Spain
 msgomez@ub.edu

Dr. N. C. Hinkle
 Professor
 Dept. of Entomology
 Univ. of Georgia
 Athens, GA 30602-2603
 NHinkle@uga.edu
 flea biology and control.

Miss Theresa M. Howard
 Collection Manager - Diptera, Siphonaptera, Arachnida & Myriapoda
 Dept. of Entomology, The Natural History Museum
 Cromwell Road
 London SW7 5BD
 UK
 t.howard@nhm.ac.uk
<http://www.nhm.ac.uk/nature-online/life/insects-spiders/index.html>
<http://www.nhm.ac.uk/research-curation/departments/entomology/index.html>
 morphological taxonomy of Siphonaptera and Diptera and the care and preservation of insect collections.

Mr. James (Jim) R. Kucera
 5930 Sultan Circle
 Murray, UT 84107-6930
 jameskucera@aol.com
 flea systematics & taxonomy, host relationships, distribution & biogeography.

Dra. Marcela Lareschi
 Centro de Estudios Parasit. Vectores
 CEPAVE (CONICET-UNLP)
 Calle 2 # 584
 La Plata (1900)
 Argentina
 mlareschi@cepave.edu.ar

Dr. Norbert Mencke
 Director Global Veterinary Services
 Bayer HealthCare AG
 Building: Mon 6210
 Leverkusen
 Germany
norbert.mencke@bayerhealthcare.com
<http://www.bayerhealthcare.com>

Dr. Barry M. OConnor
 Curator & Professor
 Museum of Zoology
 University of Michigan
 1109 Geddes Ave
 Ann Arbor, MI 48109-1079
bmoc@umich.edu
 ectoparasites of vertebrates, particularly mites.

Jeff Shryock, M.S.
 Sr. Research Biologist
 Merial, Ltd., Missouri Research Center
 6498 Jade Rd.
 Fulton, MO 65251
Jeffrey.shryock@merial.com

Dr. Andrew Smith
 Vet and Biomedical Sciences
 Murdoch University
 Murdoch 6150
 West Australia
Andrew.Smith@murdoch.edu.au
 ecology; fleas as vectors of parasites and associated diseases.

Ms. Amoret Whitaker MSc DIC
 Dept. of Entomology, The Natural History Museum
 Cromwell Road
 London, Great Britain SW7 5BD
a.whitaker@nhm.ac.uk
www.nhm.ac.uk/research-curation/staff-directory/entomology/cv-3536.html
 PhD student - forensic entomology.

Featured Research

Lareschi, Marcela, JULIANA NOTARNICOLA, SANTIAGO NAVA, AND GRACIELA NAVONE. 2007. Parasite Community (Arthropods and Filarioids) Associated with Wild Rodents from the Marshes of La Plata River, Argentina. *Comp. Parasitol.* 74(1):141-147.

We investigated the parasite community (arthropods and filarioids) associated with wild rodents from a marshy area in Buenos Aires province, Argentina. During 1995 and 1996, 105 sigmodontine rodents were trapped in The Reserva Selva Marginal de Punta Lara and in La Balandra (Argentina). Six sigmodontine species with a relative density (RDI) of 11.1% were trapped. A total of 3,371 specimens from 16 species of arthropods and 2 species of filarioids were collected. The values of parasite species richness (S) and prevalence (P) were as follows: from *Scapteromys aquaticus* S = 9; P = 96%, from *Oligoryzomys* spp. S = 8; P = 100%, from *Oxymycterus rufus* S = 8; P = 93%, and from *Akodon azarae* S = 6; P = 93%. The single *Deltamys kemp*i captured was not parasitized. Parasite communities of *S. aquaticus* and *Ox. rufus* showed the highest similarity resulting from their overlapped microhabitats, not only when presence/absence of species was considered (Sorenson's index; C_{ss} = 82%), but also when the proportion of every species was taken into account (Pianka's index; O = 26.67). The 2 species of filarioids collected were restricted to *Oligoryzomys* spp. and *Ox. rufus*. The absence of lice and filarioids from *Ox. rufus* from La Balandra suggests an isolation of this host population from the northern localities. Considering arthropod–filarioid associations, a significant affinity was observed from *Ox. rufus* from Punta Lara, between *Litomosoides oxymycteri* and the mite *Ornithonyssus bacoti*. *Scapteromys aquaticus* and *Ak. azarae* may not be susceptible to filarioids; although *Or. bacoti* were recovered from both.

Smith, A., Telfer, S., Burthe, S., Bennett, M. and Begon, M. (2005) Trypanosomes, fleas and field voles: ecological dynamics of a host-vector-parasite interaction. *Parasitology* **131**: 355-365.

To investigate the prevalence of a flea-borne protozoan (*Trypanosoma (Herpetosoma) microti*) in its field vole (*Microtus agrestis*) host, we monitored over a two-year period a range of intrinsic and extrinsic parameters pertaining to host demographics, infection status and vector (flea) prevalence. Generalized Linear Mixed Modelling was used to analyse patterns of both flea and trypanosome occurrence. Overall, males of all sizes and ages were more likely to be infested with fleas than their female counterparts. Flea prevalence also showed direct density dependence during the winter, but patterns of density dependence varied amongst body mass (age) classes during the summer. Trypanosome prevalence did not vary between the sexes but was positively related to past flea prevalence with a lag of three months, with the highest levels occurring during the autumn season. A convex age-prevalence distribution was observed, suggesting that individuals develop a degree of immunity to trypanosome infection with age and exposure. An interaction between age and whether the individual was new or recaptured suggested that infected animals are less likely to become territory holders than their uninfected counterparts.

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 simultaneous stay in the flea *Citellophilus tesquorum* at different environmental
 temperatures]
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Necrology

Harold J. Egoscue (1917-2002)

Publications (partial listing)

Hastriter, M.W., Egoscue, H.J., and R. Traub. 1998. "A description of the male of *Jordanopsylla allredi* Traub and Tipton, 1951, and characterization of the tribes within Anomiopsyllinae (Siphonaptera: Ctenophthalmidae)." *Entomological Society of Washington. Proceedings of the Entomological Society of Washington* 100, no. 1: 141-146.

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Egoscue, Harold Joseph. Ecology and life history of the kit fox in Tooele County, Utah. Salt Lake City : University of Utah, 196-

Egoscue, Harold Joseph. A new hairless mutation in deer mice. Salt Lake City : University of Utah, 196-

Egoscue, Harold Joseph. The bushy-tailed wood rat: a laboratory colony. Salt Lake City : University of Utah, 196-

Egoscue, Harold Joseph. Color phases and their inheritance in the northern grasshopper mouse. Salt Lake City : University of Utah, 196-

Egoscue, Harold Joseph. New and additional host-flea associations and distributional records of fleas from Utah. Salt Lake City : University of Utah, 196-

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Egoscue, Harold Joseph. Small mammal records from western Utah. Salt Lake City : University of Utah, 1959.

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Robert E. Elbel (1925-2005)

<http://www.alumni.utah.edu/u-news/january06/memorial.htm>

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Publications

Elbel, R.E. (1951) Comparative studies on the larvae of certain species of fleas (Siphonaptera). J. Parasit. 37:119-28, 22 figs. [*Ctenocephalides felis* (Bouché), *Nosopsyllus fasciatus* (Bosc), *Orchopeas leucopus* (Baker), *O. s. sexdentatus* (Baker)].

Elbel, R.E. (1952) Comparative morphology of some rat flea larvae (Siphonaptera). J. Parasit. 38:230-38, 14 figs. and pictorial key to 8 species. [*Echidnophaga gallinacea* (Westwood), *Xenopsylla cheopis* (Rothschild), *Polygenis gwyni* (C. Fox), *Leptopsylla segnis* (Schönherr)].

Elbel, R.E. (1954) Plague studies in Thailand. Bull. Thai Dept. Publ. Hlth 24(3):8.

Traub, R., P.T. Johnson, M.L. Messe & R.E. Elbel (1954) Isolation of *Rickettsia tsutsugamushi* from rodents from Thailand. Amer. J. Trop. Med. Hyg. 3:356-59.

Elbel, R.E. & K.C. Emerson (1956) A new mallophagan from the White Pelican. Ent. News 67:173-75, 6 figs. (*Pectinopygus tordoffi* n. sp.).

Emerson, K.C. & R.E. Elbel (1956) A new mallophagan from Southeast Asia. Ent. News 67: 117-20, 6 figs. (*Meropoecus smithi* n. sp.).

Elbel, R.E. & M. Thaineua (1957) Plague prevention by *Xenopsylla cheopis* control in Thailand. Amer. J. Trop. Med. Hyg. 6:280-93, 4 figs.

Emerson, K.C. & R.E. Elbel (1957) New records of Mallophaga from wild chickens. J. Parasit. 43:381-82.

Emerson, K.C. & R.E. Elbel (1957) New species and records of Mallophaga from Gallinaceous birds of Thailand. Proc. Ent. Soc. Wash. 59:232-43, 19 figs. (*Cuclotogaster phayrei* n. sp., *C. subinsolitus* n. sp., *Goniodes chloropus* n. sp., *Lipeurus boonsongi* n. sp., *L. deignani* n. sp., *Oxylipeurus annamensis* n. sp.).

Emerson, K.C. & R.E. Elbel (1957) A new genus of Ischnocera (Mallophaga). Fla. Ent. 40:147-48, 3 figs. (*Galliphilopterus* n. gen., *G. brunneopectus* n. sp.).

Emerson, K.C. & R.E. Elbel (1957) New species and records of *Strigiphilus* (Philopteridae: Mallophaga) from Thailand. Proc. Biol. Soc. Wash. 70:195-200, 10 figs. (*S. ketupae* n. sp., *S. siamensis* n. sp., *S. macrogenitalis* n. sp., *S. heterogenitalis* n. sp.).

Emerson, K.C. & R.E. Elbel (1957) Two new species of *Mulcticola* (Philopteridae: Mallophaga). Can. Ent. 89:420-23, 10 figs. (*M. deignani* n. sp., *M. minutus* n. sp.).

Emerson, K.C. & R.E. Elbel (1958) A new species and subgenus of *Kurodaia* (Mallophaga: Amblycera). Ent. News 69:29-32, 3 figs. [*Falcomenopon* n. subgen., *Kurodaia* (*Falcomenopon*) *boonsongi* n. sp., raised to generic rank, *F. boonsongi* Emerson & Elbel, by Price & Beer (1963, Ann. Ent. Soc. Amer. 56: 379-85)].

Taylor, E.H. & R.E. Elbel (1958) A contribution to the herpetology of Thailand. Univ. Kan. Sci. Bull. 38:1033-1189, 36 figs. (*Micrixalus magnapustulosus* n. sp., *Leiolopisma siamensis* n. sp., *Parahelicops boonsongi* n. sp., *Mabuya macularia postnasalis* n. subsp., *M. m. quadrifasciata* n. subsp., *M. m. malcolmi* n. subsp., *Sibynophis collaris triangularis* n. subsp.).

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Elbel, R.E. & K.C. Emerson (1960) A new species of *Kurodaia* (Mallophaga: Amblycera) from the Collared Scops Owl of Thailand. Proc. Biol. Soc. Wash. 73:119-22, 8 figs. (*K. deignani* n. sp.).

Coatney, G.R., R.E. Elbel & P. Kocharatana (1960) Blood parasites of birds and mammals of Loei Province, Thailand. J. Parasit. 46:701-02. (*Haemoproteus columbae* type, *H. velans* type, *Leucocytozoon* types-round and spindle, *Plasmodium* sp., *Trypanosoma lewisi* type, *Microfilariae*).

Elbel, R.E. (1961) Key for translating Thai to English. Post Mailbag, Bangkok Post 17(161), July 17. (47 to 22 consonants, 47 to 20 vowels).

Emerson, K.C. & R.E. Elbel (1961) A new species of *Rallicola* (Mallophaga) from Southeast Asia. Ent. News 72:130-32, 3 figs. (*R. indicus* n. sp.).

Elbel, R.E. (1963) End-of-Tour Report. Malaria Eradication in South

Thailand. US Dept. State AID A-2349:1-10. (malaria in 2,504,955 people reduced from 31% to 2%, goal 0.05%).

Elbel, R.E. (1967) The amblyceran Mallophaga (biting lice) found on the Bucerotidae (Hornbills). Proc. US Nat. Mus. 120:1-76, 72 figs. (*Chapinia fasciati* n. sp., *C. camuri* n. sp., *C. clayae* n. sp., *C. waniti* n. sp., *C. malayensis* n. sp., *C. hoplai* n. sp., *C. boonsongi* n. sp., *C. wenzeli* n. sp., *C. blakei* n. sp., *C. traylori* n. sp., *C. Iydae* n.sp., *C. muesebecki* n. sp., *Bucerocolpocephalum* n. gen., *B. emersoni* n. sp., *B. deignani* n. sp.).

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Elbel, R.E. (1969) *Chapinia elbeli* Tendeiro, a synonym of *Chapinia fasciati* Elbel (Mallophaga: Menoponidae). Proc. Biol. Soc. Wash. 82:489-90.

Elbel, R.E. (1969) The taxonomic position of the Hornbill *Rhyticeros plicatus subruficollis* (Blyth) as indicated by the Mallophaga. Condor 71:434-35.

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Elbel, R.E., G.T. Crane, L.E. Stipe, G.B. Van Nosedol & K.L. Smart (1971) Arbovirus isolations from mosquitoes collected at Callao, Utah, 1966 and 1967. Mosq. News 31:61-68, 1 fig. (California Group, Cache Valley viruses).

Elbel, R.E., L.T. Nielsen, D.M. Rees & F.F. Faulkner (1972) Mosquito collections for VEE virus surveillance in southern Utah. Proc. Utah Mosq. Abate. Assn. 25:25.

Smart, K.L., R.E. Elbel & G.T. Crane (1972) An isolate of Western encephalitis virus from southern Utah mosquitoes in 1972. Proc. Utah Mosq. Abate. Assn. 25:25.

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Elbel, R.E. & R.D. Price (1973) Three new Oriental and New Guinean Degeeriella (Mallophaga: Philopteridae). Pac. Ins. 15:95-101, 17 figs. (*D. quatei* n. sp., *D. storeri* n. sp., *D. traylori* n. sp.).

Elbel, R.E. (1976) *Bucerocophorus*, a new genus of ischnoceran Mallophaga from African Hornbills (Bucerotidae). Proc. Biol. Soc. Wash. 89:313-23, 11 figs. (*B. watsoni* n. sp.).

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Elbel, R.E. (1977) *Buceroemersonia*, a new genus of ischnoceran Mallophaga found on the Hornbill genus *Tockus* (Bucerotidae). Proc. Biol. Soc. Wash. 90:798-807, 9 figs. (*B. clarkei* n. sp., *B. brelihi* n. sp.).

Elbel, R.E. (1977) Two new *Buceronirmus* (Mallophaga: Philopteridae) from *Rhyticeros undulatus* and *R. plicatus* (Hornbills). Pac. Ins. 17:413-18, 8 figs. (*B. deignani* n. sp., *B. thompsoni* n. sp.).

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southwestern Utah and northwestern Arizona insects, 1972-1975. Mosq. News 37:497-507. (Western encephalitis, Main Drain, St. Louis encephalitis, Jamestown Canyon, Anopheles A Group viruses).

Crane, G.T., R.E. Elbel & C.H. Calisher (1977) Transovarial transmission of California encephalitis virus in the mosquito *Aedes dorsalis* at Blue Lake, Utah. Mosq. News 37:479-82.

Elbel, R.E. & H.E. Stark (1978) Field key to adult rabbits and rodents of the Bonneville Basin of West Central Utah. Dugway Proving Ground Tech. Rept. DPG-TR-L425P:v+1-5. (29 species).

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Elbel, R.E. (1983) Some possible hosts of mosquitoes. Proc. Utah Mosq. Abate. Assn. 36:34. (29 mammals, 5 birds).

Crane, G.T., R.E. Elbel, D.B. Francy & C.H. Calisher (1983) Arboviruses from western Utah, USA, 1967-1976. J. Med. Ent. 20:294-300. (Cache Valley, Lokern, Main Drain, California serogroup, Western encephalitis viruses).

Elbel, R.E. (1986) Twenty years of arbovirus studies in western Utah. Proc. Utah Mosq. Abate. Assn. 38-39:35-38. (10 viruses from 628,727 pooled mosquitoes, biting midges).

Elbel, R.E. (1988) Significance of larval classification of fleas (Siphonaptera) as related to the adults. Proc. Utah Mosq. Abate. Assn. 41:30-32.

Elbel, R.E., G.T. Crane & D.B. Francy (1989) Mosquitoes and viruses from western Utah in 1985. Proc. Utah Mosq. Abate. Assn. 42:13-15. (California serogroup viruses).

Elbel, R.E. & M.J. Rosenfeld (1990) Relationship between Jackrabbit populations and California serogroup viruses in western Utah. Proc. Utah Mosq. Abate. Assn. 43:1-4, 1 fig. (inverse association).

Elbel, R.E. (1991) Siphonaptera, ch. 36, 674-89, 21 figs., in Stehr, F.W. (ed.) IMMATURE INSECTS v. 2. Kendall/Hunt Publ. Co., Dubuque, IA, xvi + 975 p. (diagnosis of 52 larval species).

Elbel, R.E., G.T. Crane, C.H. Calisher & D.B. Franczy (1991) California serogroup viruses in western Utah: Serologic results in humans and rabbits and isolations from mosquitoes. *Proc. Utah Mosq. Abate. Assn.* 44:25-29. (incl. Bunyamwera serogroup viruses).

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Elbel, R.E. (1996) Arthropod camouflage and mimicry. *Proc. Int. NW Conf. Dis. Nat. Commun. Man* 51:33-35, 1 fig. (40 Arthropods).

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Elbel, R.E. & E.R. Elbel (1998) Beware of Yakima River! Charlie Drake's

search for Plague. Proc. Int. NW Conf. Dis. Nat. Commun. Man 53: 30 figs.

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